GBDS Oncology

Inferential Seamless Designs for Treatment Selection during a Phase II/III Study in Oncology

Final Presentation, December 19th 2024

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Project Information

Duration: July 01th 2024 - December 20th 2024

Master Thesis

Objective: Efficient Phase II/III MAMS design for optimal dose/drug combination.

Focus:

- Drop-the-Losers approach (Wason et al., 2017)
- Strong advantages vs other MAMS Design
- Smaller sample sizes, higher power, reduced duration

  → Cost Reduction, Time-to-Market

- Test the assumption with a simulation study

Goal: Extension of the concept to binary and survival endpoints

SMMR
STATISTICAL METHODS IN MEDICAL RESEARCH

Article

Statistical Methods in Medical Research
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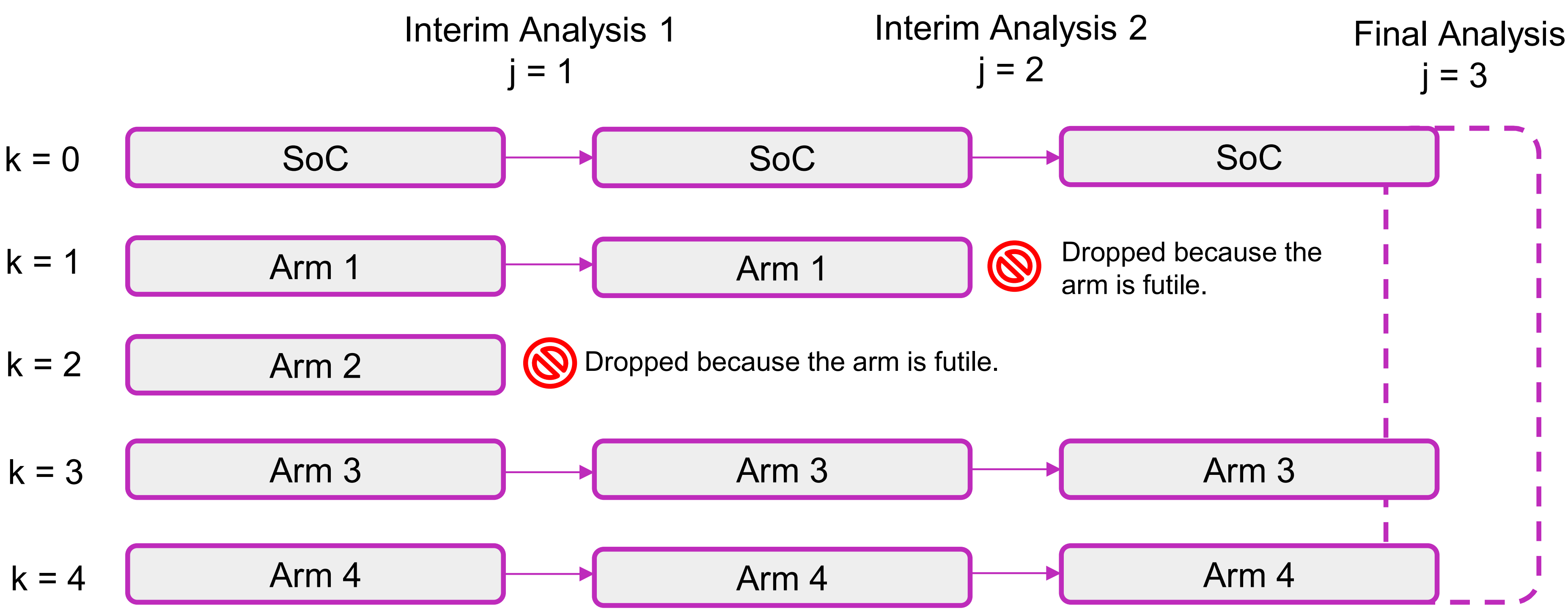
A multi-stage drop-the-losers design for multi-arm clinical trials

James Wason,[1] Nigel Stallard,[2] Jack Bowden[1] and Christopher Jennison[3]

MAMS

Definition: MAMS trials, multiple experimental treatments tested against SoC in the same trial.



Drop-the-Losers Design

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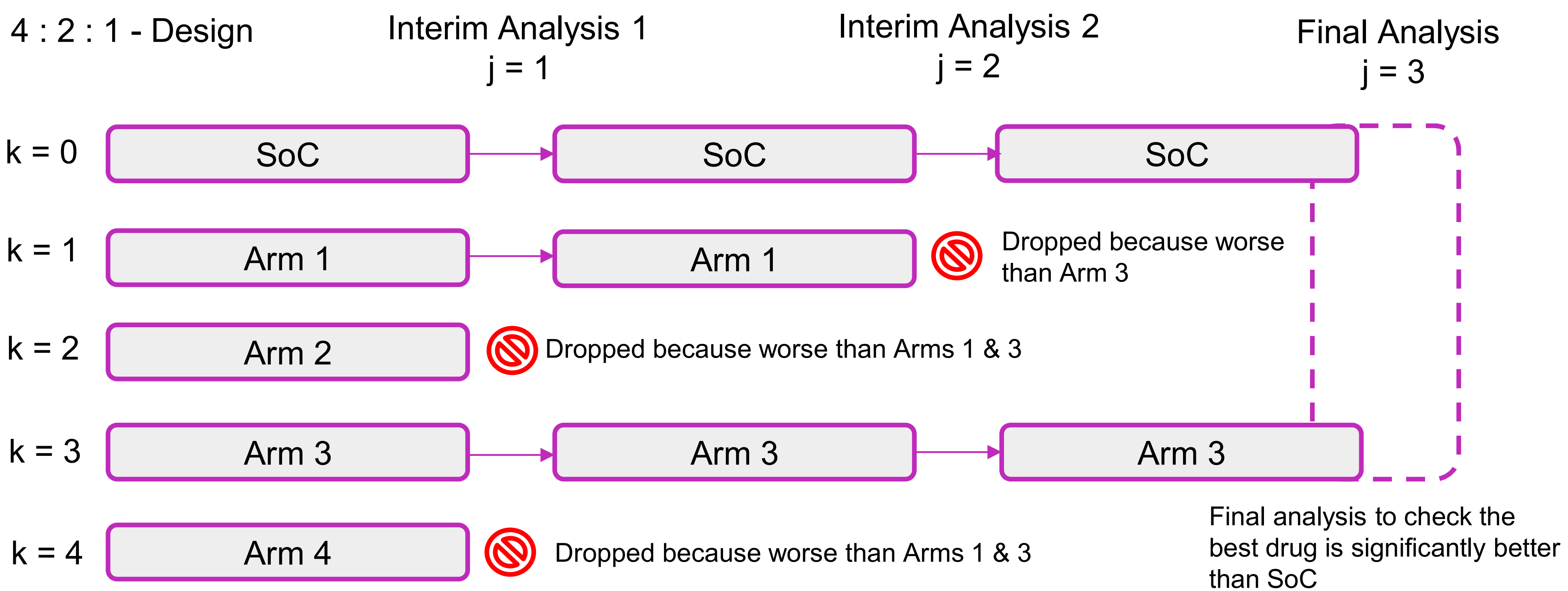

The test statistic must be greater than a specific critical value (threshold c).

Drop-the-Losers Design

Advantages:

- Study Cost & Sample Size: The sample sizes are fixed, allowing for precise cost estimation.
- Flexibility: adaptable to interim analyses and continuous monitoring, with the ability to remove ineffective arms, offering advantages over independent trials (parallel)
- Reduced administrative burden: only one placebo group used; single-arm trials require individual study approvals.
- Practical Applications: Ideal for trials with multiple arms, quick elimination of ineffective treatments, advantage over traditional designs.

Limitations:

- Computational complexity, error control challenges, difficulty with small effect sizes, complex calculations, robust planning needed.

Source: Wason et al., 2017

Derivation Covariance Matrix

Continuous Endpoints

$$Z_j^{(k)} = \left(\frac{\sum_{i=1}^{n_j} X_{ki}}{n_j} - \frac{\sum_{i=1}^{n_j} X_{0i}}{n_j}\right)\sqrt{\frac{n_j}{2\sigma^2}}$$

Assume $l > j$ and $n_l > n_j$ and $k = m$

We want to solve: $\quad \mathrm{Cov}\left(Z_j^{(k)}, Z_l^{(k)}\right)$?

$$\mathrm{Cov}(Z_j^{(k)}, Z_l^{(k)}) = \mathrm{Cov}\left(\left(\frac{\sum_{i=1}^{n_j} X_{ki}}{n_j} - \frac{\sum_{i=1}^{n_j} X_{0i}}{n_j}\right)\sqrt{\frac{n_j}{2\sigma^2}}, \left(\frac{\sum_{i=1}^{n_l} X_{ki}}{n_l} - \frac{\sum_{i=1}^{n_l} X_{0i}}{n_l}\right)\sqrt{\frac{n_l}{2\sigma^2}}\right)$$

$$= \sqrt{\frac{n_j n_l}{(2\sigma^2)^2}} \cdot \left[\mathrm{Cov}\left(\frac{\sum_{i=1}^{n_j} X_{ki}}{n_j}, \frac{\sum_{i=1}^{n_l} X_{ki}}{n_l}\right) - \mathrm{Cov}\left(\frac{\sum_{i=1}^{n_j} X_{ki}}{n_j}, \frac{\sum_{i=1}^{n_l} X_{0i}}{n_l}\right)\right.$$

$$\left. - \mathrm{Cov}\left(\frac{\sum_{i=1}^{n_j} X_{0i}}{n_j}, \frac{\sum_{i=1}^{n_l} X_{ki}}{n_l}\right) + \mathrm{Cov}\left(\frac{\sum_{i=1}^{n_j} X_{0i}}{n_j}, \frac{\sum_{i=1}^{n_l} X_{0i}}{n_l}\right)\right]$$

$$= \sqrt{\frac{n_j n_l}{2\sigma^2}}\left[\frac{1}{n_j n_l}\sum_{i=1}^{n_j}\mathrm{Cov}(X_{ki}, X_{ki}) + \frac{1}{n_j n_l}\sum_{i=1}^{n_j}\mathrm{Cov}(X_{0i}, X_{0i})\right]$$

$$= \sqrt{\frac{n_j}{n_l}}$$

$Notation:$
j & l for analysis
k & m for arms
X for Outcome of the Patient

X_{kj} : Outcomes for treatment group k at stage j.
X_{0j} : Outcomes for control group at stage j.
n_j : Total number of observations at stage j.
σ^2 : Known variance.

Source: Wason et al., 2017

Derivation Covariance Matrix

Continuous Endpoints

$$Z_j^{(k)} = \left(\frac{\sum_{i=1}^{n_j} X_{ki}}{n_j} - \frac{\sum_{i=1}^{n_j} X_{0i}}{n_j} \right) \sqrt{\frac{n_j}{2\sigma^2}}$$

Assume $l > j$ and $n_l > n_j$ and $k \neq m$

We want to solve: $\quad \text{Cov}\left(Z_j^{(k)}, Z_l^{(m)}\right)$?

$Notation:$
j & l for analysis
k & m for arms
X for Outcome of the Patient

$$\text{Cov}\left(Z_j^{(k)}, Z_l^{(m)}\right) = \frac{1}{2}\sqrt{\frac{n_j}{n_l}}$$

Source: Wason et al., 2017

Contrast Matrix A

$$
\begin{array}{c|c}
 & \begin{array}{cccc} \text{Arm 1} & \text{Arm 2} & \text{Arm 3} & \text{Arm 4} \end{array} \\
\begin{array}{c} \text{IA1} \\ \\ \text{IA2} \\ \text{FA} \\ \\ \end{array} & \mathbf{A} = \begin{pmatrix}
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 \\
1 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & -1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0
\end{pmatrix}
\end{array}
$$

$$
\mathbf{Z} = \begin{pmatrix} Z_1^1 \\ Z_2^1 \\ Z_3^1 \\ Z_1^2 \\ Z_2^2 \\ Z_3^2 \\ Z_1^3 \\ Z_2^3 \\ Z_3^3 \\ Z_1^4 \\ Z_2^4 \\ Z_3^4 \end{pmatrix}
$$

$$AZ \sim \mathrm{N}(A\mu, A\Sigma A^T)$$

Probability of a specific treatment being recommended:

$$\psi = (\psi_1, \ldots, \psi_K)$$

We introduce random variables,

where ψ_k is the ranking of treatment k.

$$P\left(\text{Reject } H_0^{(k)} | \delta\right) = P\left(\psi_k = 1, Z_3^{(k)} > c | \delta\right)$$

Ranking:

(1) the treatment that reaches the final analysis has rank 1;
(2) the treatment that is dropped at the first analysis with the lowest test statistic is given rank, K;
(3) if treatment k_1 reaches a later stage than treatment k_2, then $\psi_{k_1} < \psi_{k_2}$, that is, treatment k_1 has a higher ranking;
(4) if treatments k_1 and k_2 are dropped at the same stage, and k_1 has a higher test statistic at that stage, then $\psi_{k_1} < \psi_{k_2}$.

Source: Wason et al., 2017

Global Hypotheses

Global Null Hypothesis:
Definition: The global null hypothesis assumes that none of the treatments are more effective than the control.
Formally: $\mu_1 = \mu_2 = \mu_3 = \mu_4 = \mu_{SoC}$

Global Alternative Hypothesis:
Definition: The global alternative hypothesis posits that at least one treatment is significantly better than the control.
Formally: $H_A\ \exists\ i\ such\ that\ \mu_i > \mu_{SoC}$

Expected mean effect: $\mathbf{m(\delta)}$
The expected mean effect m(δ) represents the anticipated average treatment effect for each arm compared to the control group.

Source: Wason et al., 2017

Type I Error & Power

Probability of recommending any treatment under the global null hypothesis:

Under global null hypothesis H_0: each element $m(\delta)$= 0. Symmetry: equal probability of ordering ψ and Z > c. Probability of treatment recommendation

$$K\,!\,\mathbb{P}\Big(\psi_1 = 1, \psi_2 = 2, \ldots, \psi_K = K, Z_J^{(1)} > c|\delta = 0\Big)$$

Probability of recommending a specific treatment under the least favorable configuration:

$$(K-1)!\,\mathbb{P}\Big(\psi_1 = 1, \psi_2 = 2, \ldots, \psi_K = K, Z_J^{(1)} > c|\delta_1 = \delta^{(1)}, \delta_2 = \delta^{(0)}, \ldots, \delta_K = \delta^{(0)}\Big)$$

Source: Wason et al., 2017

Distribution of the Test Statistics

$$Z_j^{(3)} = \left(\frac{\sum_{i=1}^{n_j} X_{3i}}{n_j} - \frac{\sum_{i=1}^{n_j} X_{0i}}{n_j} \right) \sqrt{\frac{n_j}{2\sigma^2}}$$

X_{kj} : Outcomes for treatment group k at stage j.
X_{0j} : Outcomes for control group at stage j.
n_j : Total number of observations at stage j.
σ^2 : Known variance.

$$Z_j^{(3)} \sim N(\delta_3 \sqrt{\frac{n_j}{2\sigma^2}}, 1)$$

$\delta_3 = \mu_3 - \mu_0$
The null hypotheses to be tested are $H_0^3 : \delta_3 \leq 0$.
The global null hypothesis, H_G, is defined as $H_G : \delta_1 = \delta_2 = \ldots \delta_k = 0$

Source: Wason et al., 2017

Extension Survival Endpoints

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Survival Endpoints

$$Z_j^{(k)} \approx \frac{\log \widehat{HR}_{kj}}{\sqrt{\frac{4}{d_j^{(o)}+d_j^{(k)}}}} = \log \widehat{HR}_{kj} \cdot \sqrt{\frac{d_j^{(o)}+d_j^{(k)}}{4}}$$

Assuming $k = m$

$l > j \quad (n_l > n_j)$

$$\operatorname{Cov}\left(Z_j^{(k)}, Z_l^{(k)}\right) = \frac{1}{4}\operatorname{Cov}\left(\log \widehat{HR}_{kj} \times \sqrt{d_j^{(0)} + d_j^{(k)}}; \log \widehat{HR}_{kl} \times \sqrt{d_l^{(0)} + d_l^{(k)}}\right)$$

$$\widehat{HR}_{kl} \approx \widehat{HR}_{kj} \times \left(\frac{d_j^{(0)} + d_j^{(k)}}{d_l^{(0)} + d_l^{(k)}}\right)^{\left(1 - \frac{d_j^{(0)}+d_j^{(k)}}{d_l^{(0)}+d_l^{(k)}}\right)}$$

$$\log(\widehat{HR}_{kl}) \approx \left(\frac{d_j^{(0)} + d_j^{(k)}}{d_l^{(0)} + d_l^{(k)}}\right)\log(\widehat{HR}_{kj}) + \left(1 - \frac{d_j^{(0)} + d_j^{(k)}}{d_l^{(0)} + d_l^{(k)}}\right)\log(\widehat{HR}_{\overline{kj}})$$

$$\operatorname{Cov}\left(Z_j^{(k)}, Z_l^{(k)}\right) =$$

$$= \frac{1}{4} \times \operatorname{Cov}\left(\log \widehat{HR}_{kj}, \log \widehat{HR}_{kj}\right) \times \frac{\left(d_j^{(0)} + d_j^{(k)}\right)^{3/2}}{\left(d_l^{(0)} + d_l^{(k)}\right)^{1/2}}$$

$$= \frac{1}{4} \times 4 \times \frac{\left(d_j^{(0)} + d_j^{(k)}\right)}{d_j^{(0)} + d_j^{(k)}} \times \sqrt{\frac{d_j^{(0)} + d_j^{(k)}}{d_l^{(0)} + d_l^{(k)}}}$$

$$= \sqrt{\frac{d_j^{(0)} + d_j^{(k)}}{d_l^{(0)} + d_l^{(k)}}}$$

$Z_j^{(k)}$: Test statistic for arm k at stage j

$\log \widehat{HR}_{kj}$: The logarithm of the estimated hazard ratio (HR) for arm k compared to the control at stage j.

$d_j^{(0)}$ and $d_j^{(k)}$: Number of observed events (deaths or failures) in the control and treatment arm k at stage j, respectively.

Simulation Study

Objective:

- Effectiveness of "Drop-the-losers" design in Phase II/III oncology trials.
- Type I error rates and statistical power in selecting promising treatments.
- Potential biases under normal distribution assumptions for test statistics.

Scenarios:

- Null Hypothesis: No effect is present.
- Hybrid 1: One effective drug.
- Hybrid 2: Two effective drugs.
- Hybrid 1 + 1: One effective drug and one drug with half the effect size.
- Harmful: One drug is harmful.
- All Effective: All drugs show an effect.

Statistical methods to be evaluated:

- Survival Analysis: Cox Regression, Log-Rank Test.
- [Binary Outcomes: Difference of Proportions, Fisher’s Exact Test]

Simulation Study: Survival I

ADEMP

Design Setup (Input)
- Dropout Rate: 10%
- Recruitment Rate: 15 patients / month
- Randomization Block Size: 6

- Control Median PFS: 8 months
- Number of Treatments 4

Drop-the-losers Design: Phase II
- Stage 1: 50 events
- Stage 2: 70 events
- Stage 3: 90 events

Drop-the-losers Design: Phase III
- Stage 1: 100 events
- Stage 2: 200 events
- Stage 3: 400 events

Simulation Setup:
- Number of Simulations: 1’000’000
  max. MCSE = 0.00112

1. Null Hypothesis: No effect
2. Hybrid 1: One effective drug
3. Hybrid 2: Two effective drugs
4. Hybrid 1 + 1: One effective, one half-effective
5. Harmful Treatment
6. All Effective: All drugs show an effect.

```
# Endpoint simulation (PFS)
PFS <- ifelse(trt == 0, rexp(length(recruit), log(2) / medianPFS_control),
              rexp(n = length(recruit), as.numeric(log(2) / medianPFS_control * Results[i, trt + 2])))
```

Parallel Simulation Loop:
- Simulations run in parallel using the package *foreach* loop
- Simulates recruitment, randomization, and endpoint generation
- Censoring

Futility Analysis:
- Calculates HR, test statistic, p-value for each treatment arm.

Futility Analysis and or drop the worst test statistics

Cox-Regression Model:

$$h(t|X) = h_0(\mathrm{t}) * e^{(\beta_1 * X_1)}$$

```
coxph(Surv(ad_PFS, event_PFS) ~ ad_trt
```

Log Rank Test:

$$\chi^2 = \frac{(O_1 - E_1)^2}{V}$$

```
surfdiff(Surv(ad_PFS, event_PFS) ~ ad_trt
```

Burton et al. (2006), Morris et al. (2019)

Simulation Study: Survival II

Only with Cox Regression:

```
Results_sim <- foreach(i = 1:Nsim, .combine = 'rbind', .packages = c('survival'), .inorder = FALSE) %dopar% {
        Results$Sim <- i
        simulate_trial(i, thresholds = c(30, 50, 100), Results = Results, RecruitmentRate = 15,
        patients_stage1 = 150, patients_stage2 = 150, block_size = 6, K = 4, medianPFS_control = 8,
        dropout_PFS= 0.1)}
```

Combined Cox Regression with Log Rank Test:

```
Results_sim <- foreach(i = 1:Nsim, .combine = 'rbind', .packages = c('survival'), .inorder = FALSE) %dopar% {
        Results$Sim <- i
        simulate_trial_log_rank(i, thresholds = c(30, 50, 100), Results = Results, RecruitmentRate = 15,
        patients_stage1 = 150, patients_stage2 = 150, block_size = 6, K = 4, medianPFS_control = 8,
        dropout_PFS = 0.1)}
```

GitHub GitHub - pfistem/BMS_Masterthesis

Simulation Study: Survival III

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Calculation Critical Value c

Set Up the Integral Function:
- Use the integral_typeIerror() function to compute the Type I error.

Input Parameters:
- c: Critical value (to be found).
- mean: Mean vector under the null hypothesis.
- var: Covariance matrix.
- K: Number of treatment arms.
- Desired family-wise error rate (FWER).

Use Numerical Root-Finding:
Find c by solving for when the integral equals the desired FWER using uniroot().

Function:
```
finddesign_survival(J = 3, K = 4, ns = c(4, 2, 1), delta1 = -log(0.7), delta0 = 0, requiredfwer = 0.05,
                    requiredpower = 0.8, groupsizes = c(1, 2/3, 5/3), events = events)
```

$c = 2.038$ ($\alpha = 0.05$) | $c = 2.338$ ($\alpha = 0.025$)

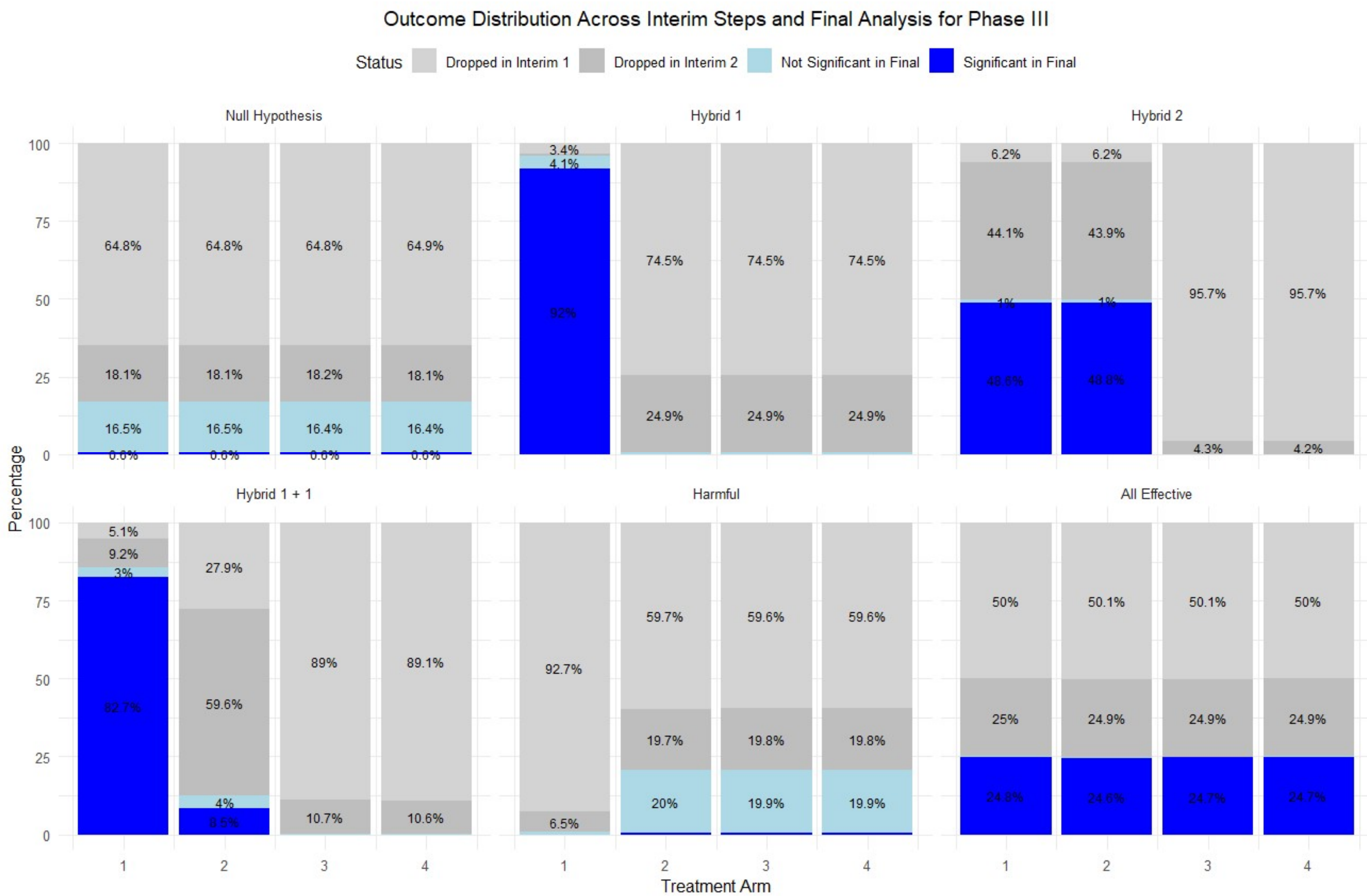
Simulation Study: Survival IV

Decision based on the HR or Test statistic

Test statistics from the Cox Regression Model and from the Log Rank Test

Scenario	Treatment Arm	Simulation	AnalysisType	Time Analysis	Outcome	HR	TestStatistics Cox	TestStatistic LogRank	Dropped
1	1	1	Interim 1	27.21366	Continue	0.833278	-0.90153	0.814976	FALSE
1	2	1	Interim 1	26.75233	Dropped due to Drop-the-Loser Design	0.944502	-0.56912	0.324246	TRUE
1	3	1	Interim 1	27.08181	Dropped due to Drop-the-Loser Design	0.963637	-0.55226	0.305297	TRUE
1	4	1	Interim 1	25.36767	Continue	0.96619	-0.67872	0.461367	FALSE
1	1	1	Interim 2	40.3487	Dropped due to Futility	1.02849	-0.28425	0.047682	TRUE
1	4	1	Interim 2	41.54692	Dropped due to Drop-the-Loser Design	0.988072	-0.33736	0.113831	FALSE
1	4	1	Final Analysis	43.4358	Continue	0.947042	-0.38425	0.147682	FALSE

Results: Survival I



FWER = 2.33 %

0.58% (0.56% - 0.59%)
0.59% (0.57% - 0.6%)
0.58% (0.56% - 0.59%)
0.58% (0.56% - 0.59%)

Values smaller than 1% are omitted for readability, except in the null hypothesis plot.

Results: Ethical Aspects

Interim Analysis 2

Scenario	Number of Arms	Total Number of Arms	Probability
1	670368	670368	1.000
2	746307	751793	0.993
3	42481	965778	0.044
4	213026	901268	0.236
5	603767	603767	1.000
6	0	997452	0.000

Final Analysis

Scenario	Number of Arms	Total Number of Arms	Probability
1	681672	681672	1.000
2	18805	979423	0.0192
3	638	996637	0.000640
4	6375	988456	0.00645
5	626802	626802	1.000
6	0	999625	0.000

Discussion

Advancing Clinical Trial Design: Extending "Drop-the-losers" Approach

Key Contribution:

- Extends "Drop-the-losers" to binary and survival endpoints for Phase II/III oncology trials.
- Addresses real-world complexities while maintaining statistical rigor.
- Integrates multiple endpoint types and uses futility analysis (and group sequential design) for earlier decision-making.
- Accelerates treatment selection and drug approval.

Limitations:

- Dynamic endpoint updates (ORR → PFS → OS) add complexity, that correlation at each endpoint's surrogacy needs to be carefully managed.

Ethical Impact:

- Minimizes exposure to ineffective treatments, aligning with beneficence.

References

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Many thanks to Pierre and Malgorzata for their support!

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Thank you very much for your attention!

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Q&A

Backup Slides

Results from Wason Paper

Table 1. Sample sizes required for a one-stage design and two-stage and three-stage drop-the-losers designs with $\alpha = 0.05$, $\beta = 0.1$, $\delta^{(1)} = 0.545$ and $\delta^{(0)} = 0.178$.

	Total sample size required for 90% power			Percentage reduction in sample size	
K	$J=1$	$J=2$	$J=3$	$J=1$ to $J=2$	$J=2$ to $J=3$
3	312	282	270	9.6	4.2
4	420	364	330	13.3	9.3
6	637	531	455	16.6	14.3
8	864	715	585	17.2	18.2

Note: For each three-stage design, the number of treatments proceeding to stage 2 is chosen to give the lowest total sample size: in the notation of Section 2, these designs are 3:2:1 for $K=3$, 4:2:1 for $K=4$, 6:3:1 for $K=6$ and 8:3:1 for $K=8$.

Source: Wason et al., 2017

Results from Wason Paper

Drop-the-Losers

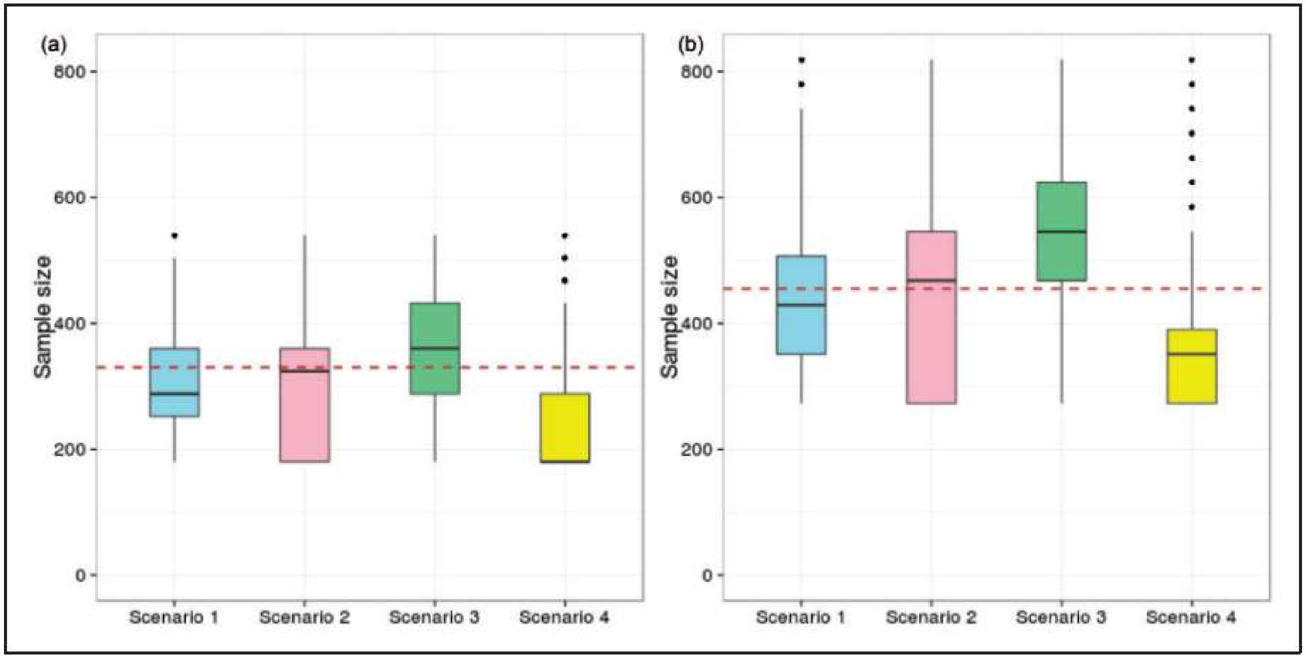


Figure 1. Sample size distribution for three-stage group-sequential MAMS designs with $K = 4$ and $K = 6$ and four vectors of treatment effects. Scenario 1 – the global null hypothesis (H_G); scenario 2 – the LFC; Scenario 3 – all experimental treatments have uninteresting treatment effect $\delta^{(0)}$; Scenario 4 – all experimental treatments have effect $-\delta^{(0)}$. The dashed red line gives the required sample size for the three-stage drop-the-losers design with the same parameters used: $\alpha = 0.05$, $\beta = 0.1$, $\delta^{(1)} = 0.545$, $\delta^{(0)} = 0.178$.

Source: Wason et al., 2017

Extension Binary Endpoints

Binary Endpoints

$$Z_j^{(k)} = \left(\frac{\sum_{i=1}^{n_j} X_{ki}}{n_j} - \frac{\sum_{i=1}^{n_j} X_{0i}}{n_j} \right) \sqrt{\frac{n_j}{2\pi(1-\pi)}}$$

$$\text{Assume } l > j \text{ and } n_l > n_j$$

$$\text{We want to solve: Cov}\left(Z_j^{(k)}; Z_l^{(k)}\right)?$$

$$k = m: \quad \text{Cov}\left(Z_j^{(k)}, Z_l^{(k)}\right) = \sqrt{\frac{n_j}{n_l}}$$

$$k \neq m: \quad \text{Cov}\left(Z_j^{(k)}, Z_l^{(m)}\right) = \frac{1}{2} \cdot \sqrt{\frac{n_j}{n_l}}$$

$Notation:$
j & l for analysis
k & m for arms
X for Outcome of the Patient

Extension Survival Endpoints

Survival Endpoints

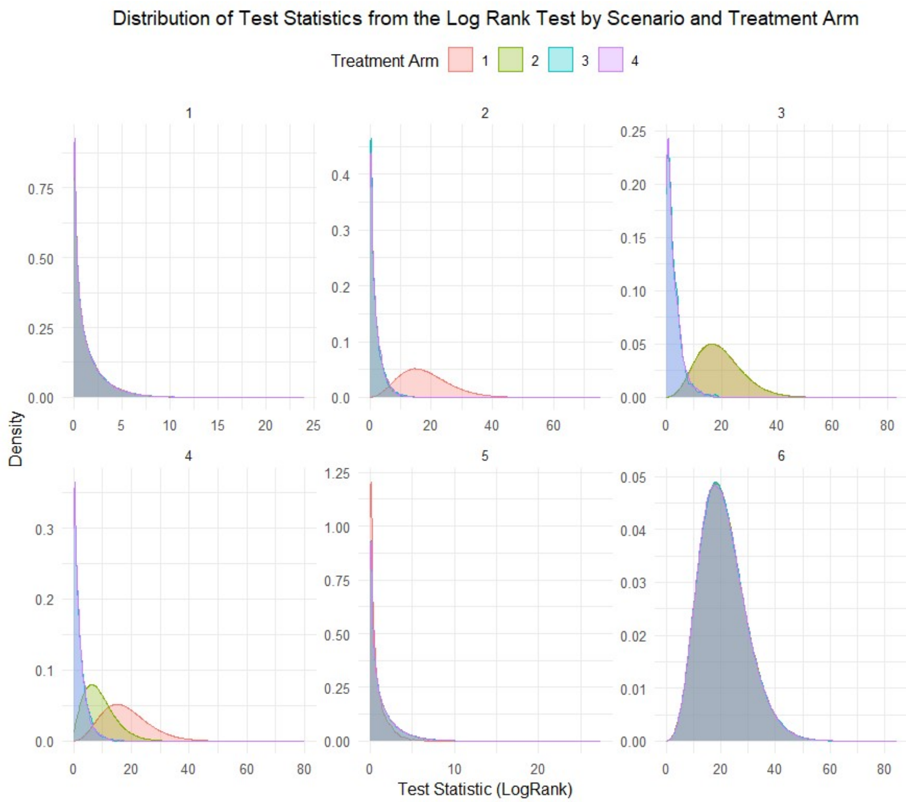
$Assuming\ k \neq m$
$p > j\ (n_p > n_j)$

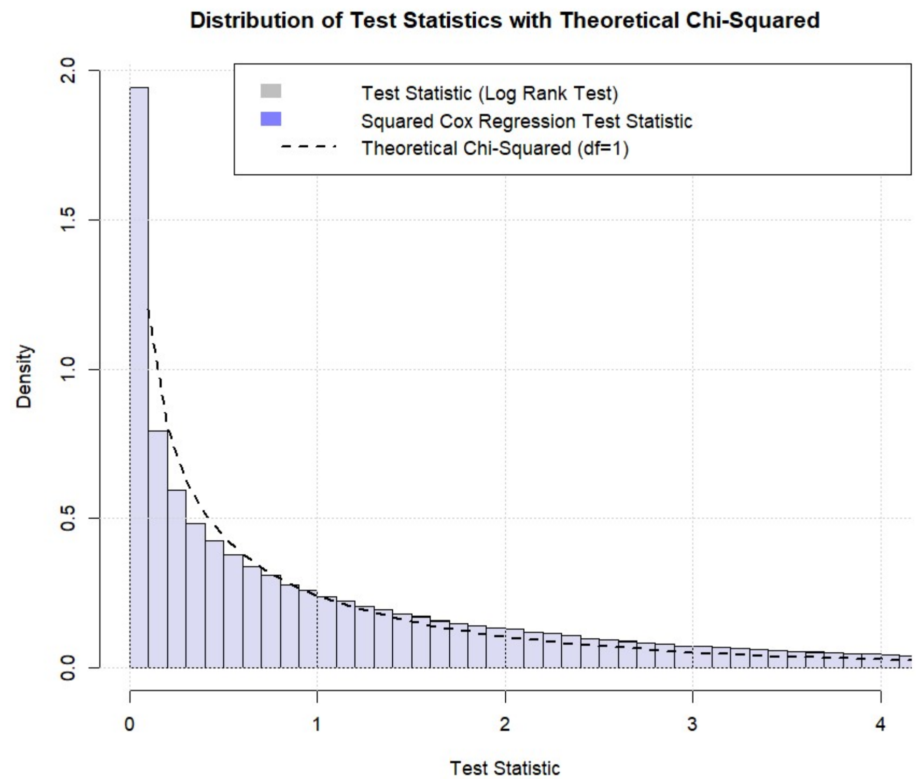
$$Cov\left(Z_j^{(k)}, Z_p^{(k)}\right) = 0.5\sqrt{\frac{d_j^{(o)} + d_j^{(k)}}{d_p^{(o)} + d_p^{(k)}}}$$

Results: Survival II

Scenario	Treatment Arm	Dropped in Interim 1	Dropped in Interim 2	Not Significant in Final (95%-CI)	Significant in Final (95%-CI)
1	1	648280 (64.83%)	181359 (18.14%)	16.46% (16.38% - 16.53%)	0.58% (0.56% - 0.59%)
1	2	648027 (64.8%)	180713 (18.07%)	16.54% (16.47% - 16.61%)	0.59% (0.57% - 0.6%)
1	3	647689 (64.77%)	182224 (18.22%)	16.43% (16.36% - 16.5%)	0.58% (0.56% - 0.59%)
1	4	648563 (64.86%)	181473 (18.15%)	16.42% (16.35% - 16.49%)	0.58% (0.56% - 0.59%)
2	1	33595 (3.36%)	5787 (0.58%)	4.1% (4.06% - 4.14%)	91.96% (91.91% - 92.02%)
2	2	744647 (74.46%)	249046 (24.9%)	0.59% (0.58% - 0.61%)	0.04% (0.04% - 0.04%)
2	3	744712 (74.47%)	249087 (24.91%)	0.58% (0.57% - 0.6%)	0.04% (0.03% - 0.04%)
2	4	744730 (74.47%)	248973 (24.9%)	0.59% (0.58% - 0.61%)	0.04% (0.03% - 0.04%)
3	1	61799 (6.18%)	441290 (44.13%)	1.05% (1.03% - 1.07%)	48.64% (48.55% - 48.74%)
3	2	62132 (6.21%)	439388 (43.94%)	1.04% (1.02% - 1.06%)	48.81% (48.71% - 48.91%)
3	3	956709 (95.67%)	42683 (4.27%)	0.05% (0.05% - 0.06%)	0.01% (0.01% - 0.01%)
3	4	956881 (95.69%)	42481 (4.25%)	0.06% (0.05% - 0.06%)	0.01% (0.01% - 0.01%)
4	1	50628 (5.06%)	92278 (9.23%)	3.03% (2.99% - 3.06%)	82.68% (82.61% - 82.76%)
4	2	278586 (27.86%)	596427 (59.64%)	4.01% (3.97% - 4.05%)	8.49% (8.43% - 8.54%)
4	3	889943 (88.99%)	106840 (10.68%)	0.29% (0.28% - 0.3%)	0.03% (0.03% - 0.03%)
4	4	890561 (89.06%)	106281 (10.63%)	0.29% (0.28% - 0.3%)	0.02% (0.02% - 0.03%)
5	1	926554 (92.66%)	64615 (6.46%)	0.88% (0.86% - 0.9%)	0% (0% - 0%)
5	2	596703 (59.67%)	197169 (19.72%)	19.96% (19.88% - 20.04%)	0.65% (0.64% - 0.67%)
5	3	595633 (59.56%)	198425 (19.84%)	19.93% (19.85% - 20.01%)	0.67% (0.65% - 0.68%)
5	4	596159 (59.62%)	197940 (19.79%)	19.94% (19.86% - 20.02%)	0.65% (0.64% - 0.67%)
6	1	499903 (49.99%)	249545 (24.95%)	0.28% (0.27% - 0.29%)	24.78% (24.69% - 24.86%)
6	2	501302 (50.13%)	249445 (24.94%)	0.28% (0.27% - 0.29%)	24.65% (24.56% - 24.73%)
6	3	501227 (50.12%)	248982 (24.9%)	0.28% (0.27% - 0.29%)	24.7% (24.62% - 24.79%)
6	4	500484 (50.05%)	249487 (24.95%)	0.27% (0.26% - 0.28%)	24.73% (24.65% - 24.82%)

Results: Survival III





Extension: Group Sequential Design

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Arm 1 Arm 2 Arm 3 Arm 4

$$\text{Futility}\left\{\begin{bmatrix} 1 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & -1 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}\right\} \begin{array}{l} > 0 \\ > 0 \\ > 0 \\ > -c_{Futility} \end{array}$$

The study should be stopped for futility if it falls below the futility threshold.

Arm 1 best at Interim Analysis 1

$$\alpha_1 \quad \begin{array}{r} \text{Futility}\left\{\begin{bmatrix} 1 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}\right\} \\ \text{GSD}\left\{\begin{matrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{matrix}\right\} \end{array} \begin{array}{l} > 0 \\ > 0 \\ > 0 \\ > c_{Futility} \\ > c_{IA1} \end{array}$$

We will not stop for futility because the test statistic for the best arm was above the threshold.

Extension: Group Sequential Design

Arm 1 best at Interim Analysis 2

$$\alpha_2 \begin{cases} \left\{ \begin{array}{rrrrrrrrrrrr} 1 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right\} & \begin{array}{l} > 0 \\ > 0 \\ > 0 \\ > c_{Futility} \end{array} \\ \left\{ \begin{array}{rrrrrrrrrrrr} -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right\} & \begin{array}{l} > -c_{IA1} \\ > 0 \\ > c_{IA2} \end{array} \end{cases}$$

The test statistic for arm 1 was neither futile nor significant –> proceed to IA2.

Final Analysis

$$\alpha_3 \begin{cases} \left\{ \begin{array}{rrrrrrrrrrrr} 1 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right\} & \begin{array}{l} > 0 \\ > 0 \\ > 0 \\ > c_{Futility} \end{array} \\ \left\{ \begin{array}{rrrrrrrrrrrr} -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right\} & \begin{array}{l} > -c_{IA1} \\ > 0 \\ > -c_{IA2} \\ > c_{Final} \end{array} \end{cases}$$

Alpha Spending Function

$$\alpha = \alpha_1 + \alpha_2 + \alpha_3$$

Test whether the test statistic for arm 1 exceeds c_{Final}.